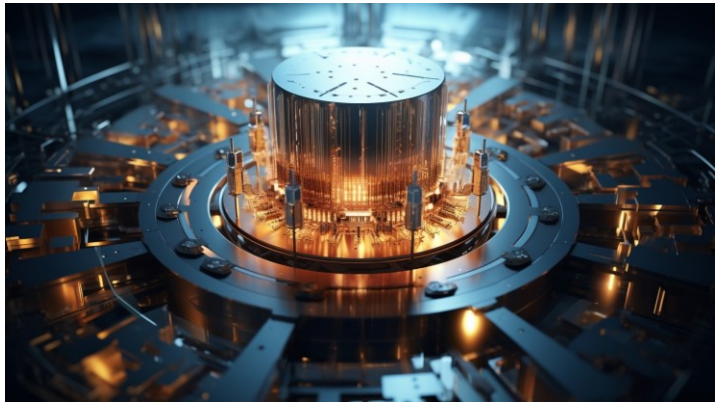
第5.3节 训练神经网络的一些细节

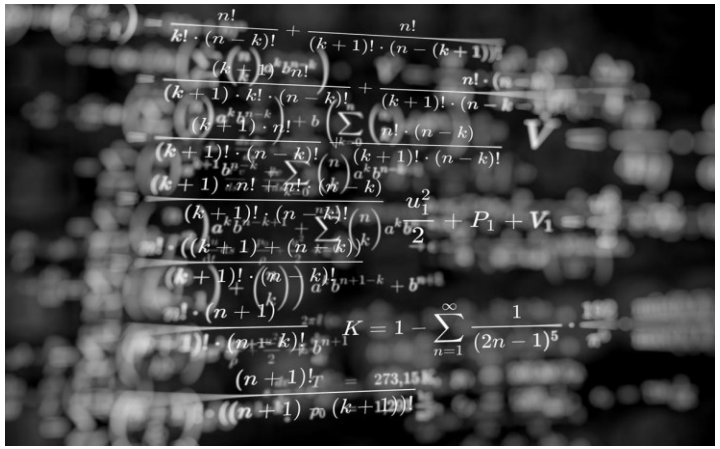
过拟合 & 梯度消失 & ReLU & Adam

当你在设计深度学习算法时……



路人眼中的你

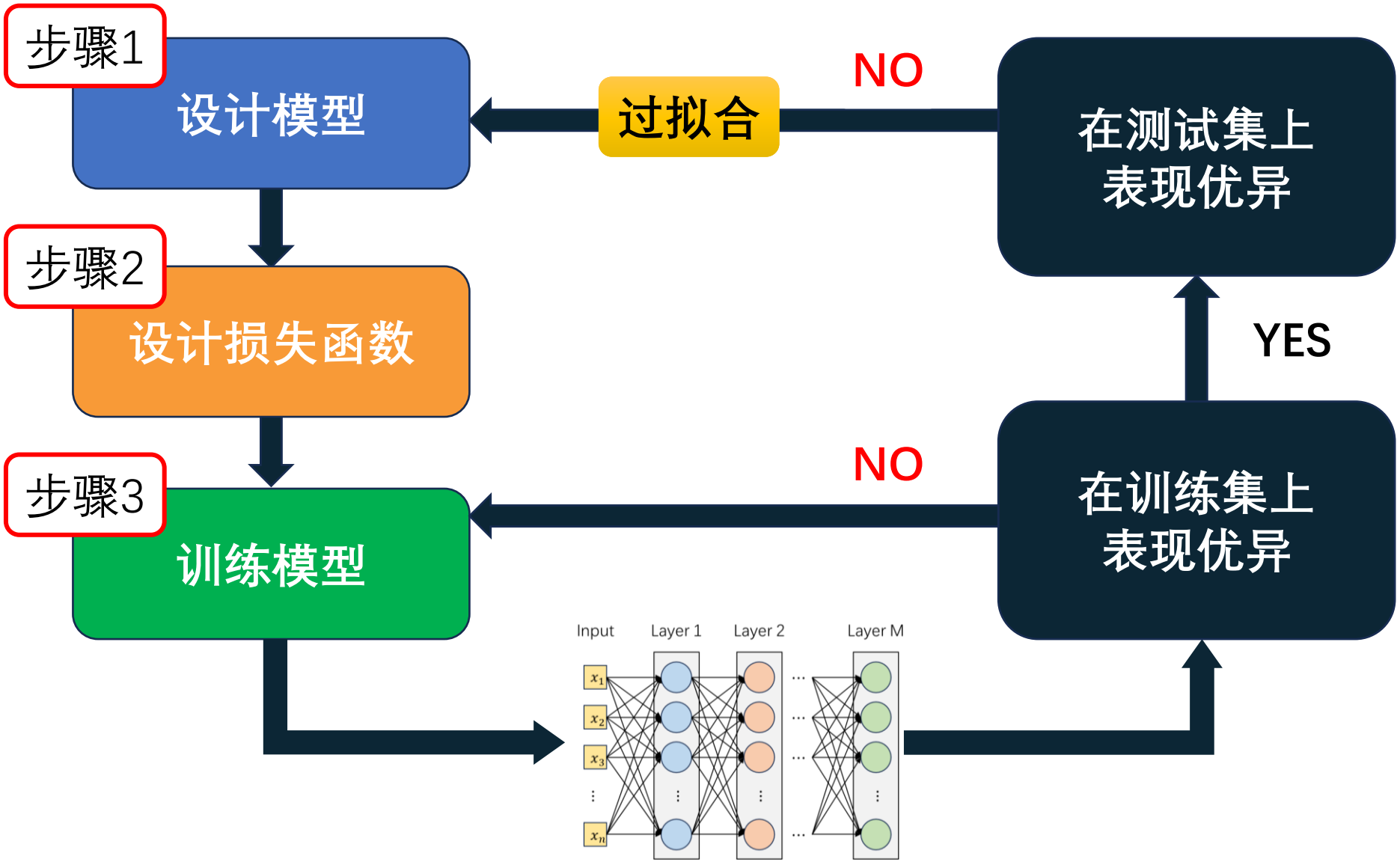
父母眼中的你



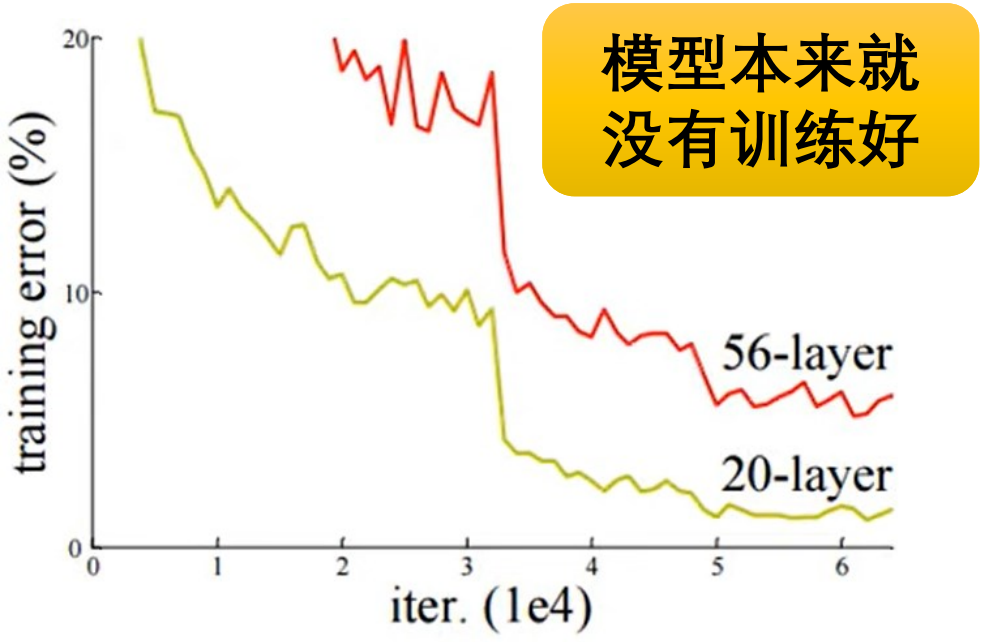
你想象中的事情

实际上做的事情

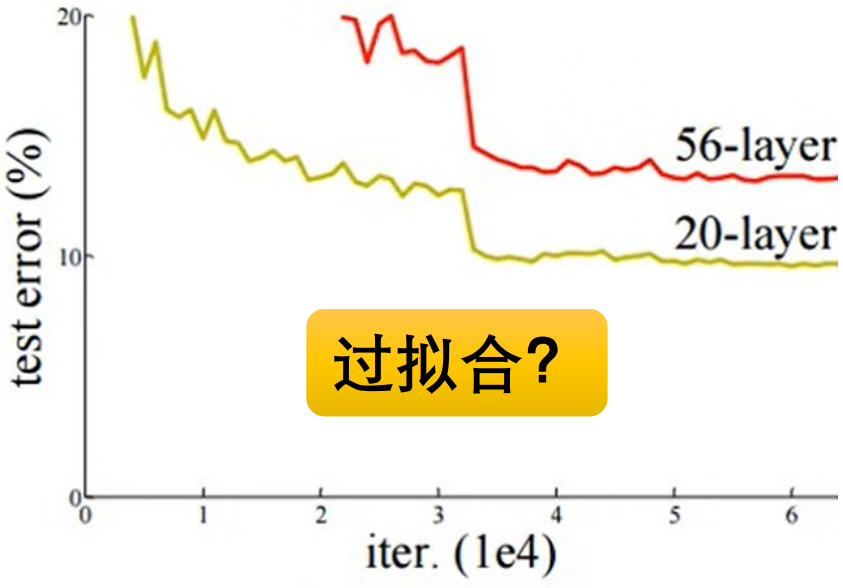
训练深度学习模型的步骤



深度学习确实发生过拟合了吗？

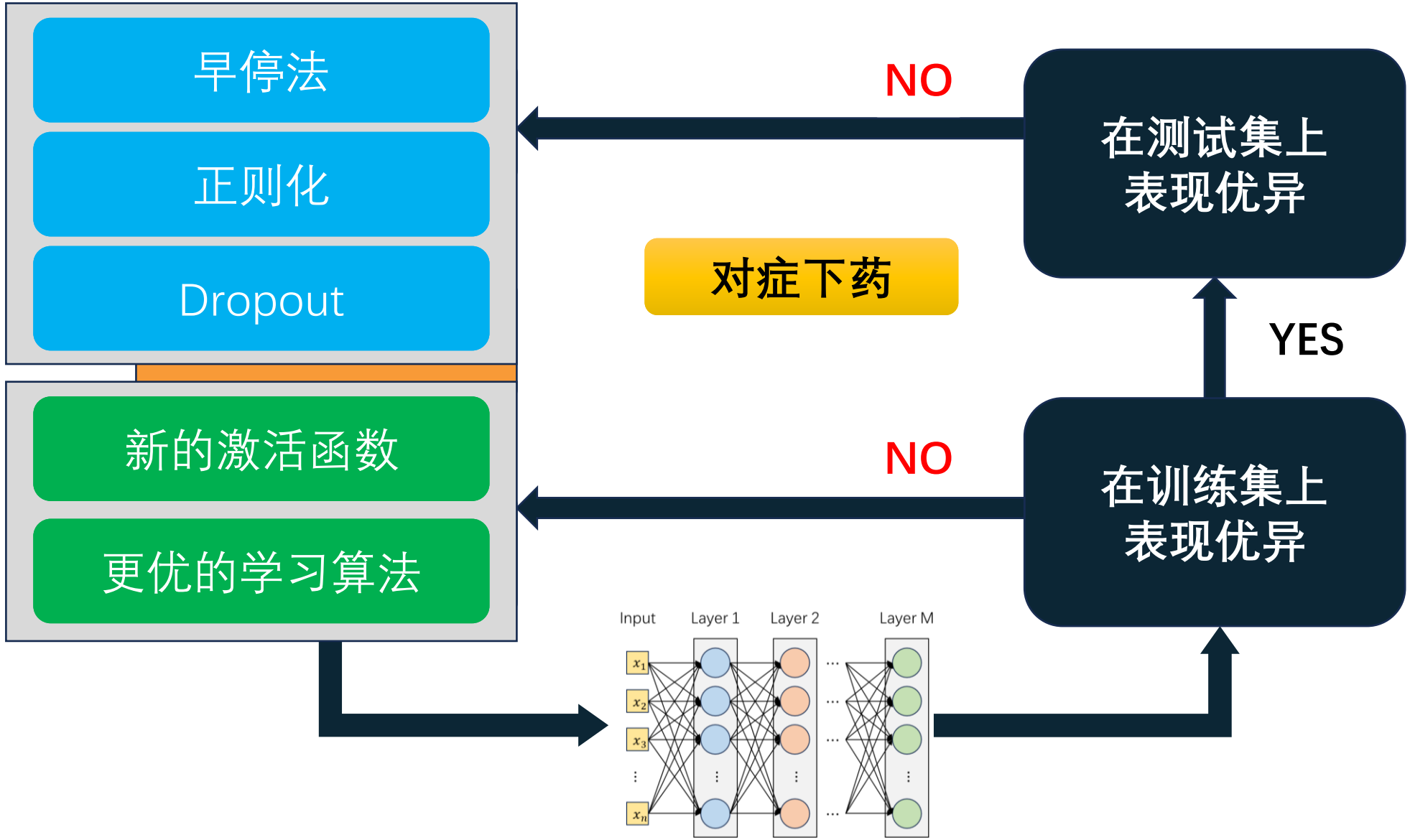


Training Data

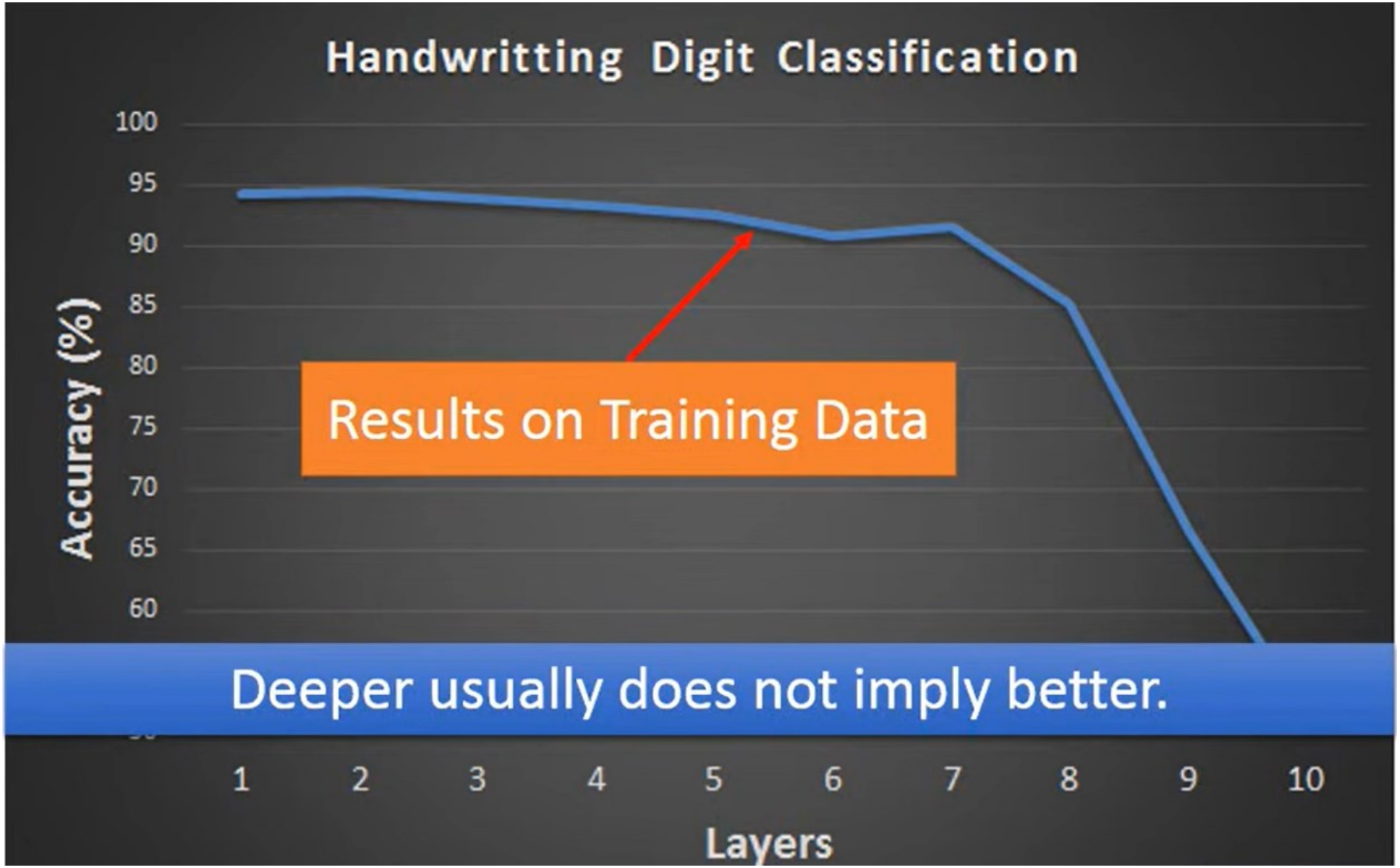


Testing Data

训练深度学习模型的步骤

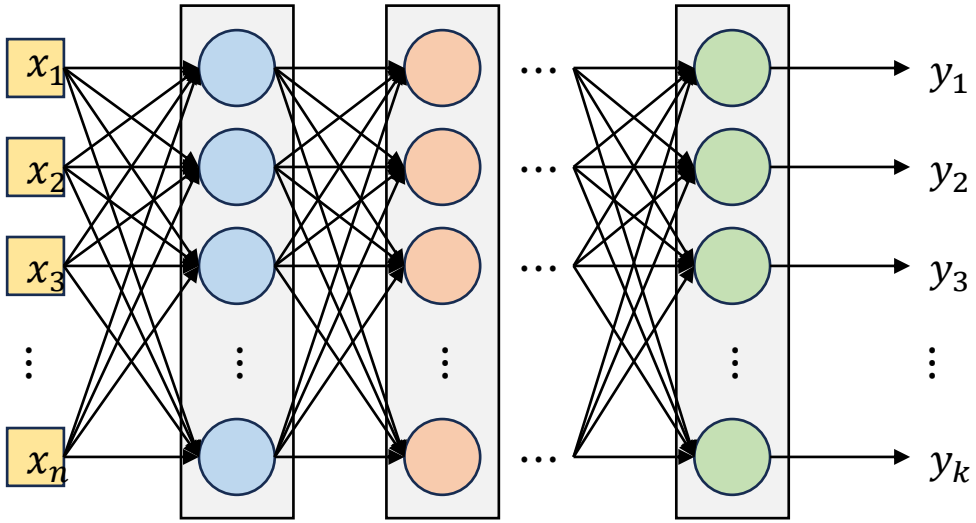


梯度消失问题



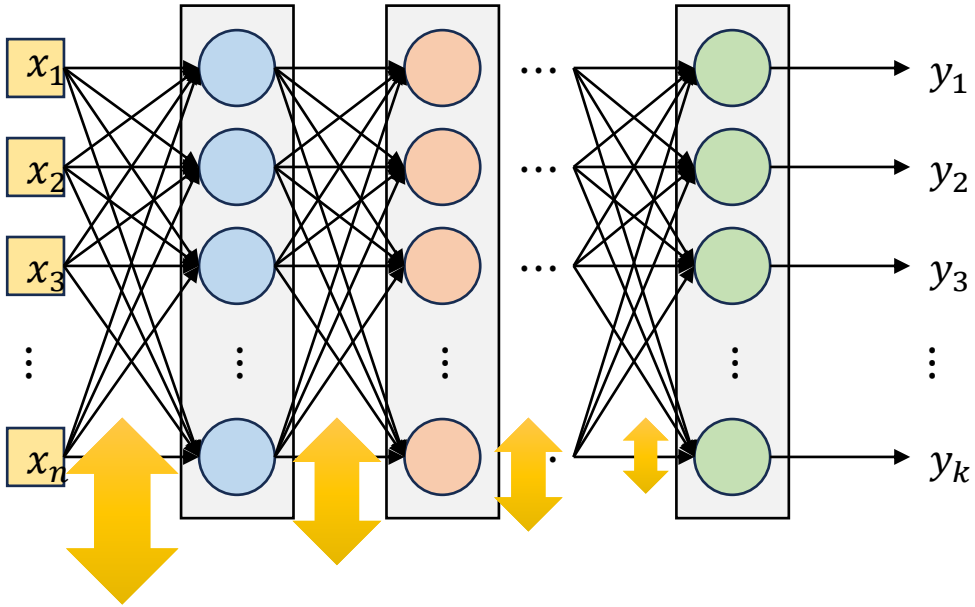
https://youtu.be/xki61j7z-30?si=Hd6awEwQb7SsSAvV

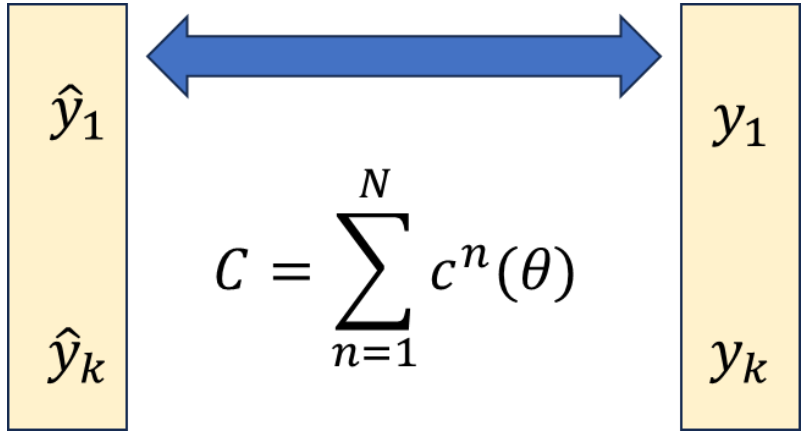
梯度消失问题



梯度较小	梯度较大
参数更新慢	参数更新快
参数处于随机状态	参数已经完成收敛

梯度消失问题

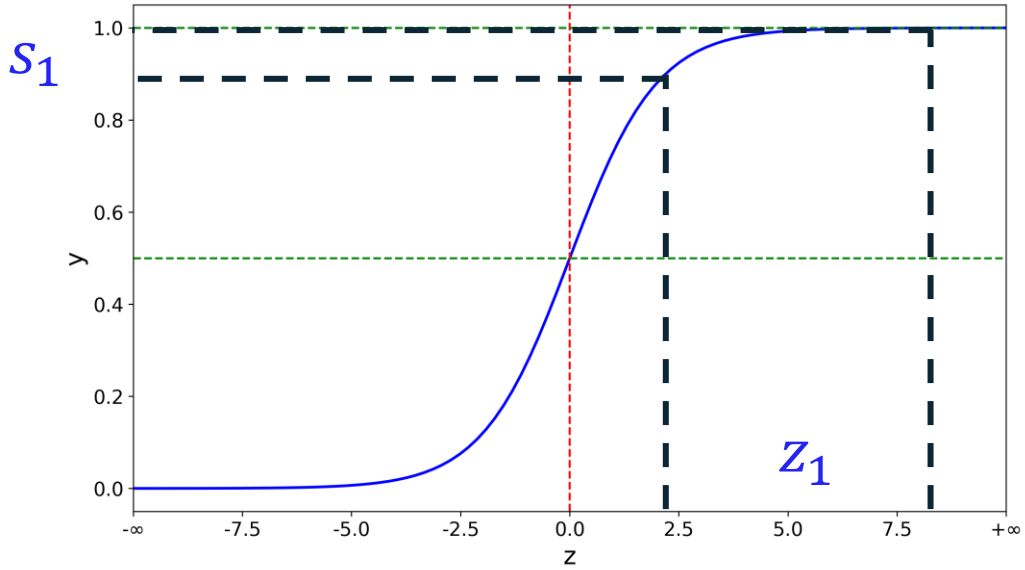




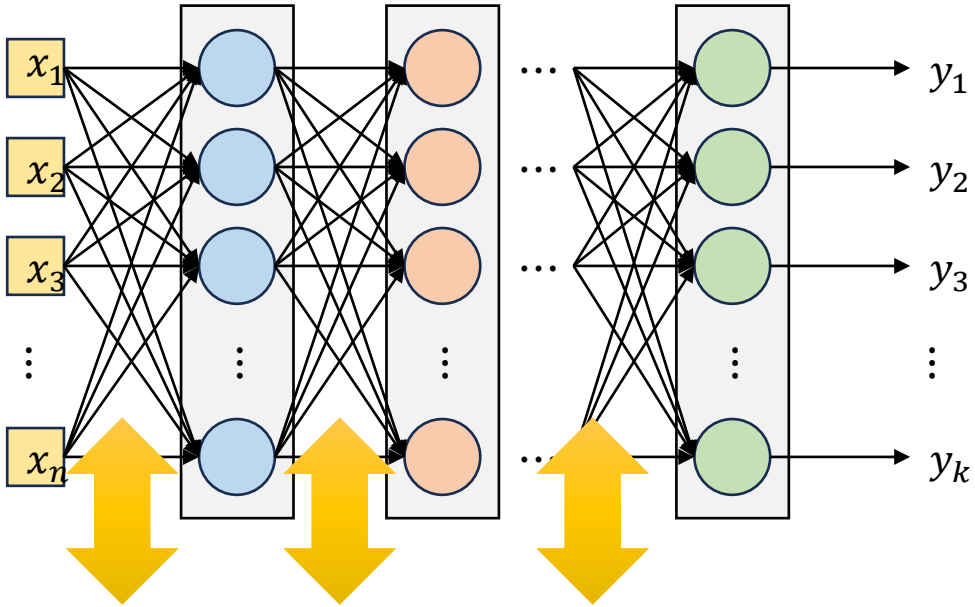
$$C = \sum_{n=1}^{N} c^n(\theta)$$

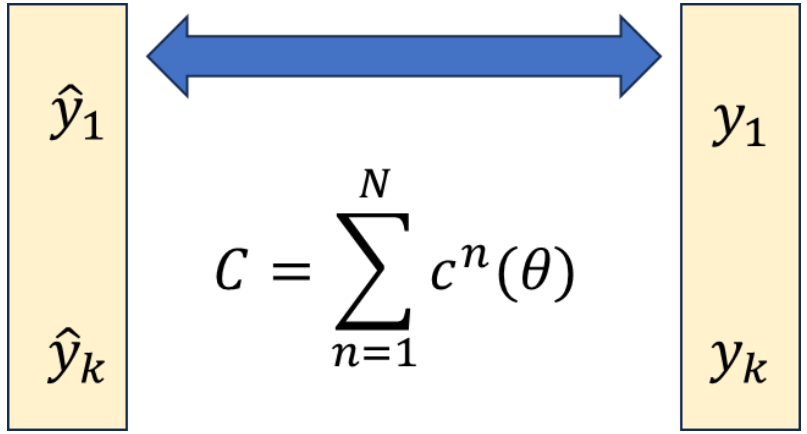
随着层数增高，低层参数对C的影响逐渐降低

$$\frac{\partial C}{\partial w} \rightarrow \frac{\Delta C}{\Delta w}$$



梯度消失问题的应对——ReLU





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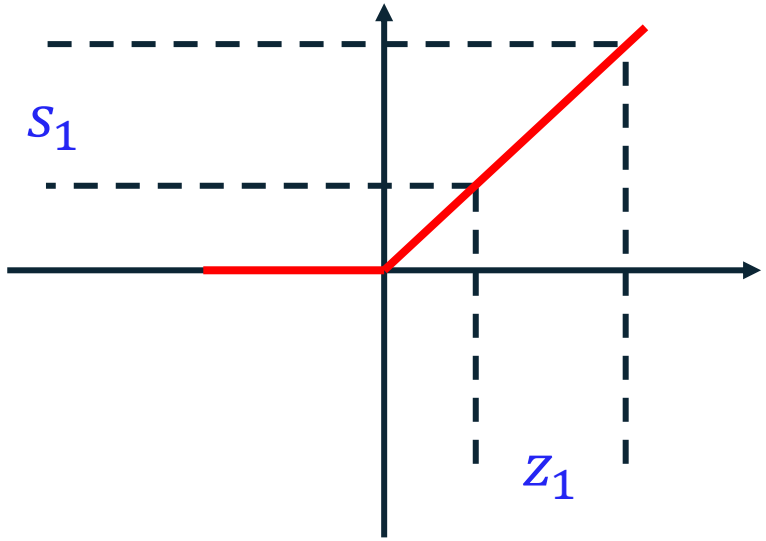
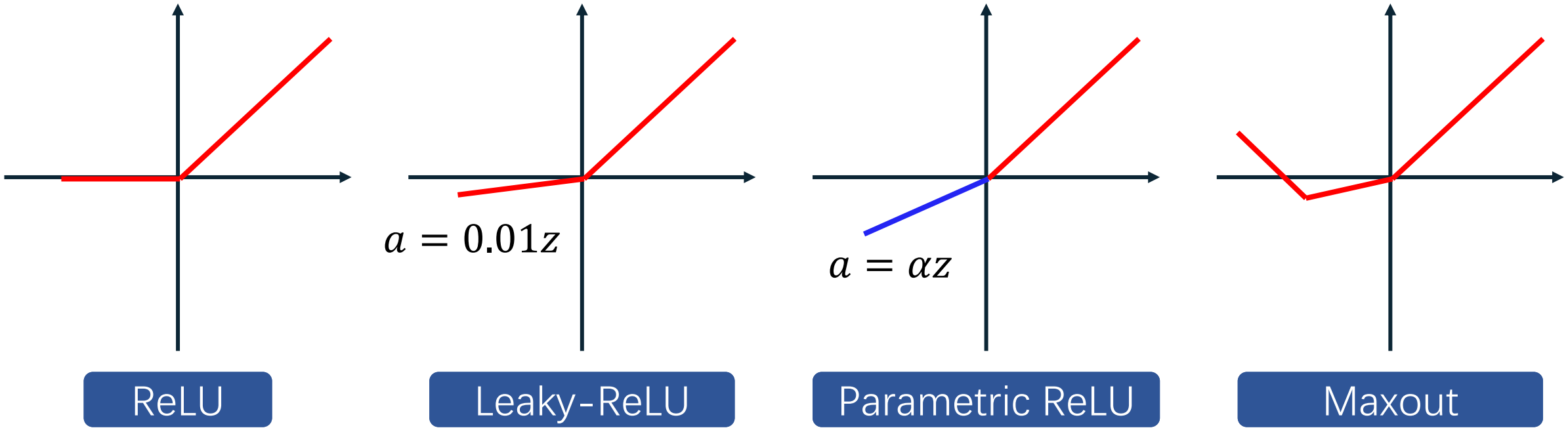
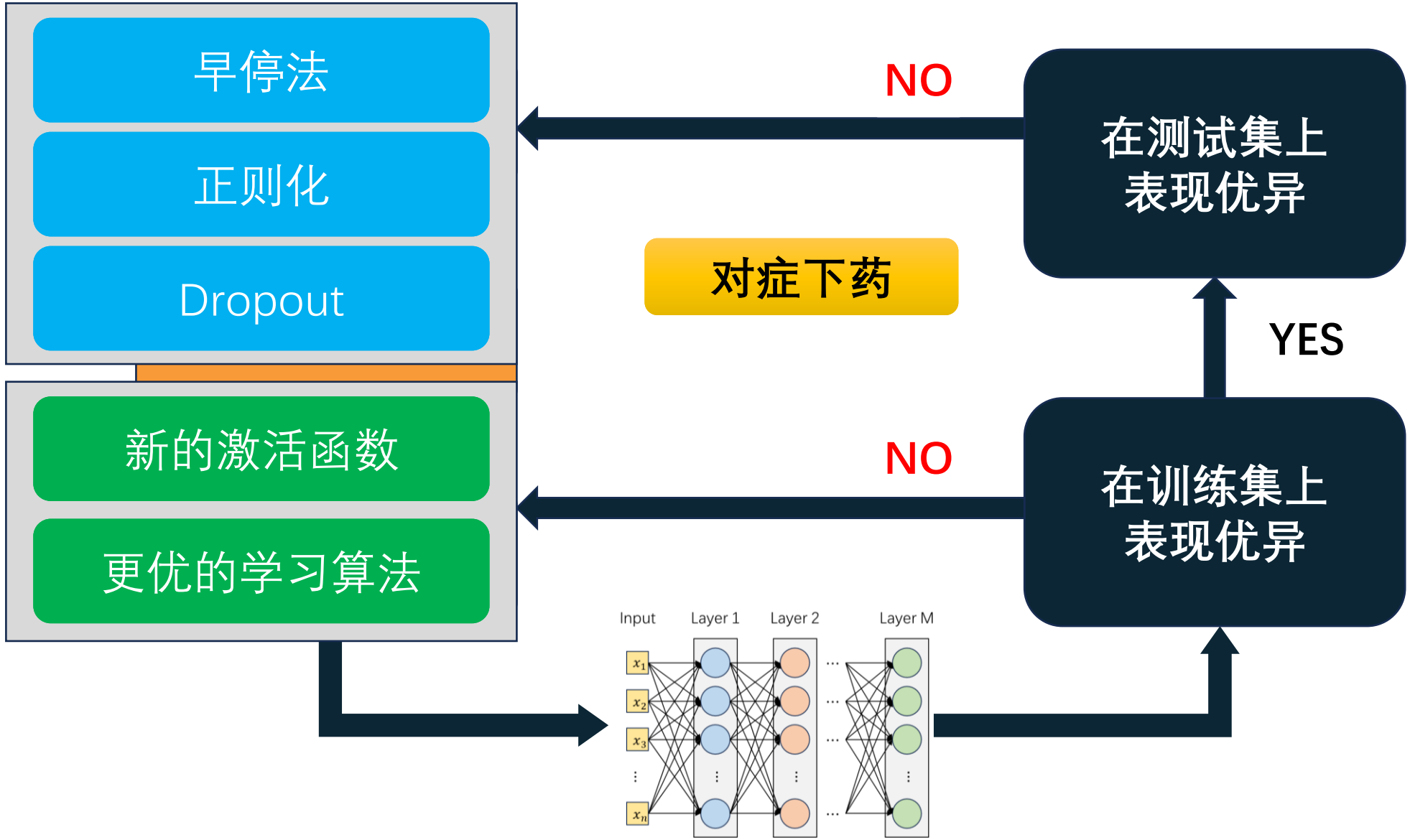


梯度消失问题的应对——ReLUs



训练深度学习模型的步骤



Adam

梯度下降步骤:

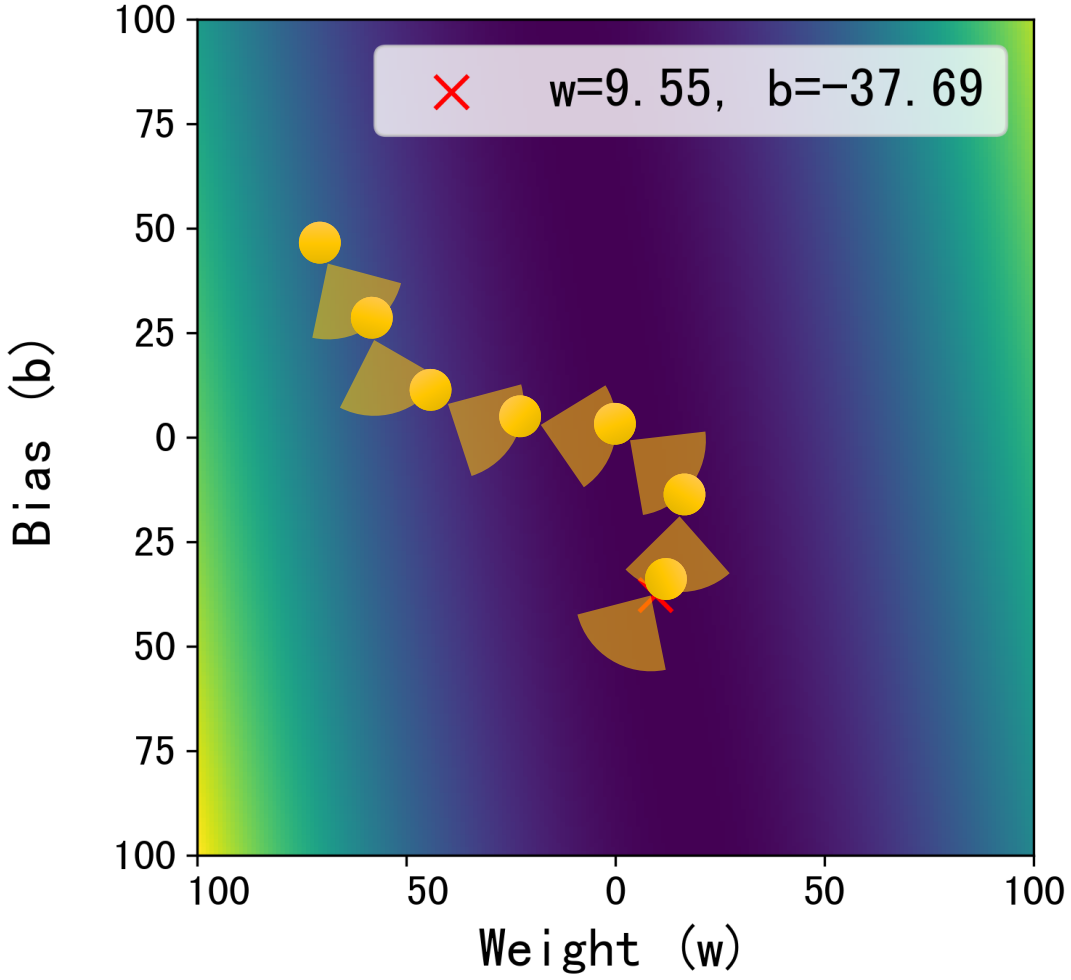
Step 1: 计算各参数的偏导数 $\frac{\partial L}{\partial b}$, $\frac{\partial L}{\partial w_i}$

Step 2: 分别更新各参数

$$b = b - \eta \frac{\partial L}{\partial b}$$

$$w_i = w_i - \eta \frac{\partial L}{\partial w_i}$$

Step 3: 不断重复 Step 1 和 Step 2



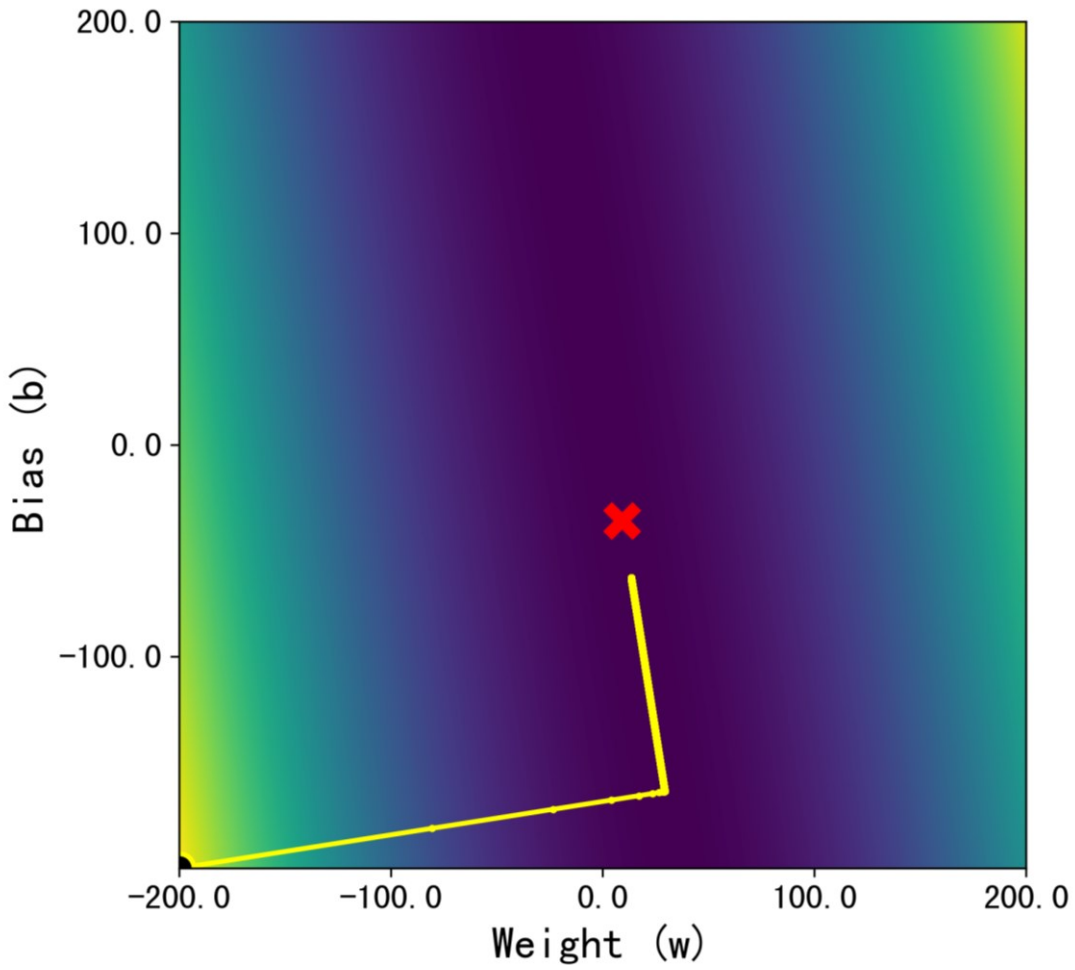
Adam

统一学习率的问题

学习w较快，学习b较慢

如果w和b的学习率同步减小，则b可能难以收敛至最优

最好根据参数定制学习率衰减方式



Adam

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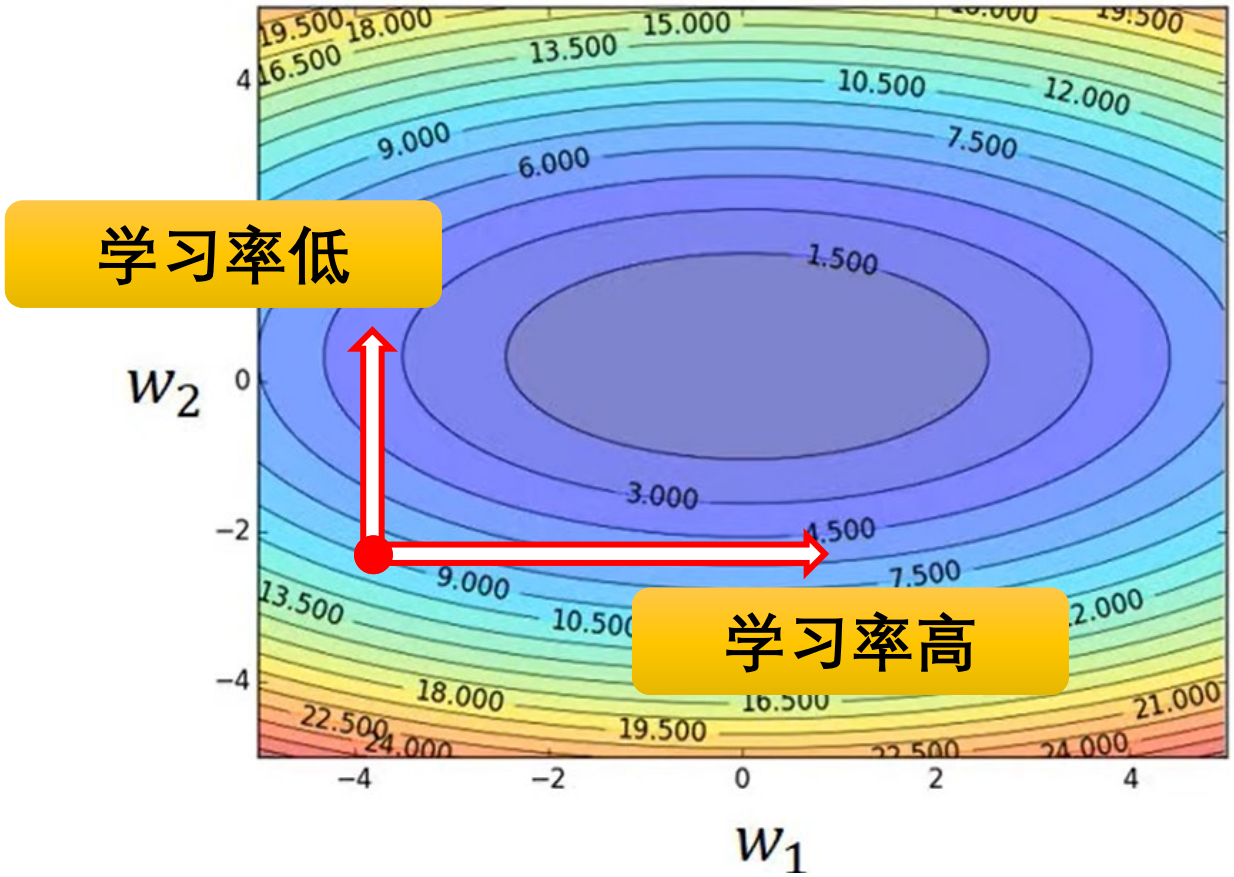
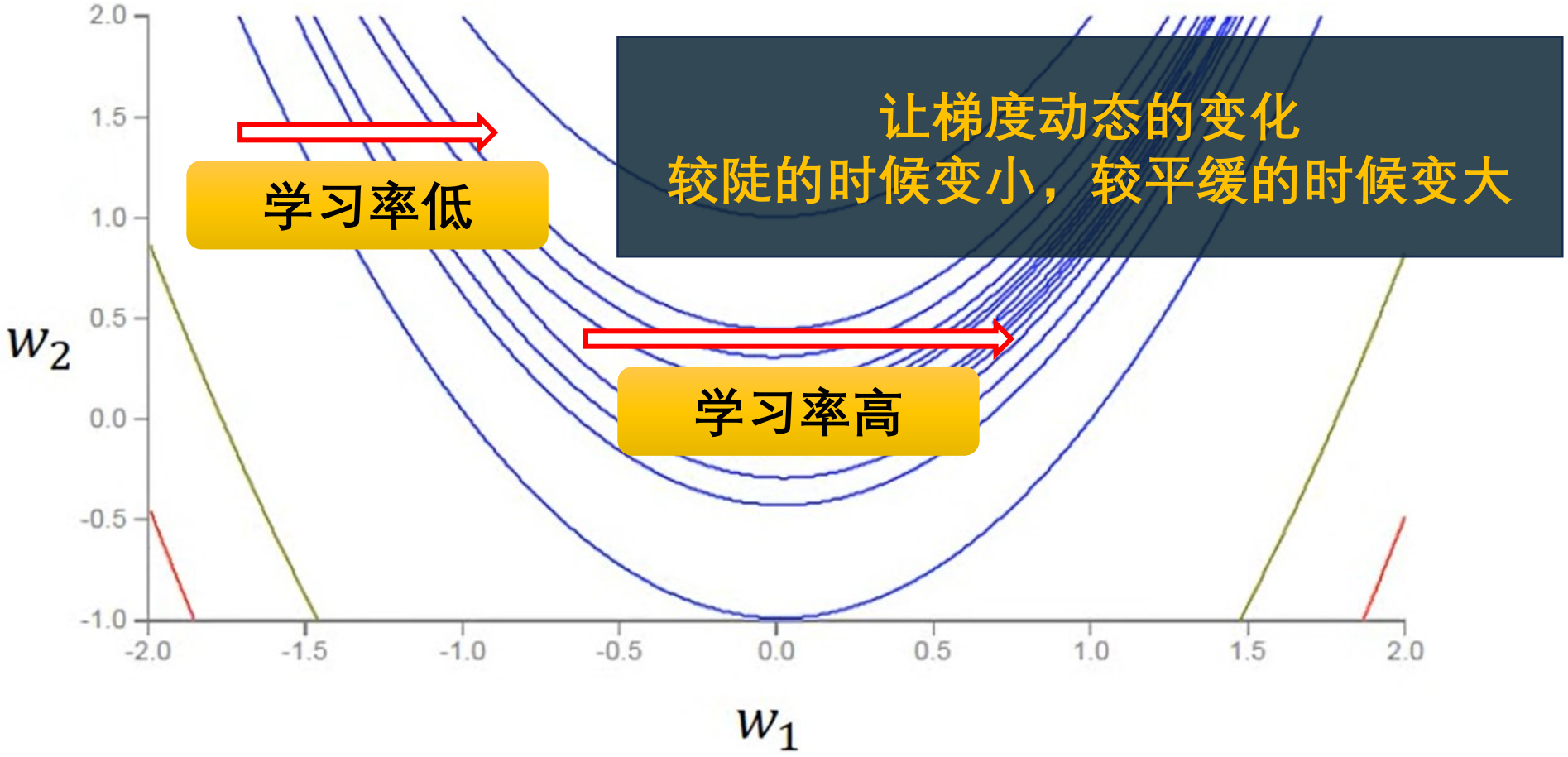


Adam

实际情况可能更加复杂



让梯度动态的变化
较陡的时候变小，较平缓的时候变大

Adam

$$w^1 \leftarrow w^0 - \frac{\eta}{\sigma^0} g^0 \qquad \sigma^0 = g^0$$

$$w^2 \leftarrow w^1 - \frac{\eta}{\sigma^1} g^1 \qquad \sigma^1 = \sqrt{\alpha(\sigma^0)^2 + (1-\alpha)(g^1)^2}$$

$$w^3 \leftarrow w^2 - \frac{\eta}{\sigma^2} g^2 \qquad \sigma^2 = \sqrt{\alpha(\sigma^1)^2 + (1-\alpha)(g^2)^2}$$

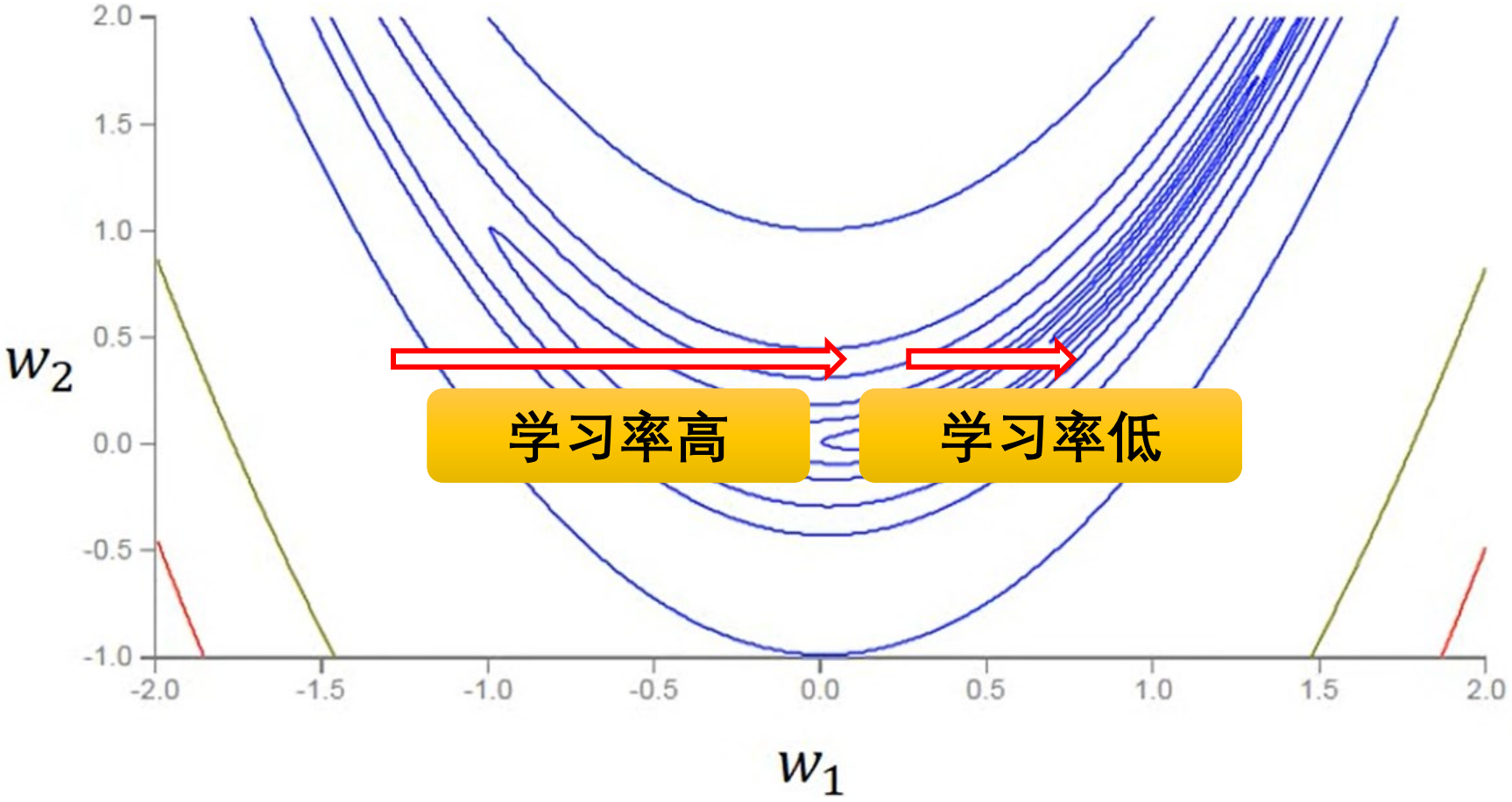
$$\vdots$$

以前的梯度　　当前的梯度

$$w^{t+1} \leftarrow w^t - \frac{\eta}{\sigma^t} g^t \qquad \sigma^t = \sqrt{\alpha(\sigma^{t-1})^2 + (1-\alpha)(g^t)^2}$$

Adam

RMSProp



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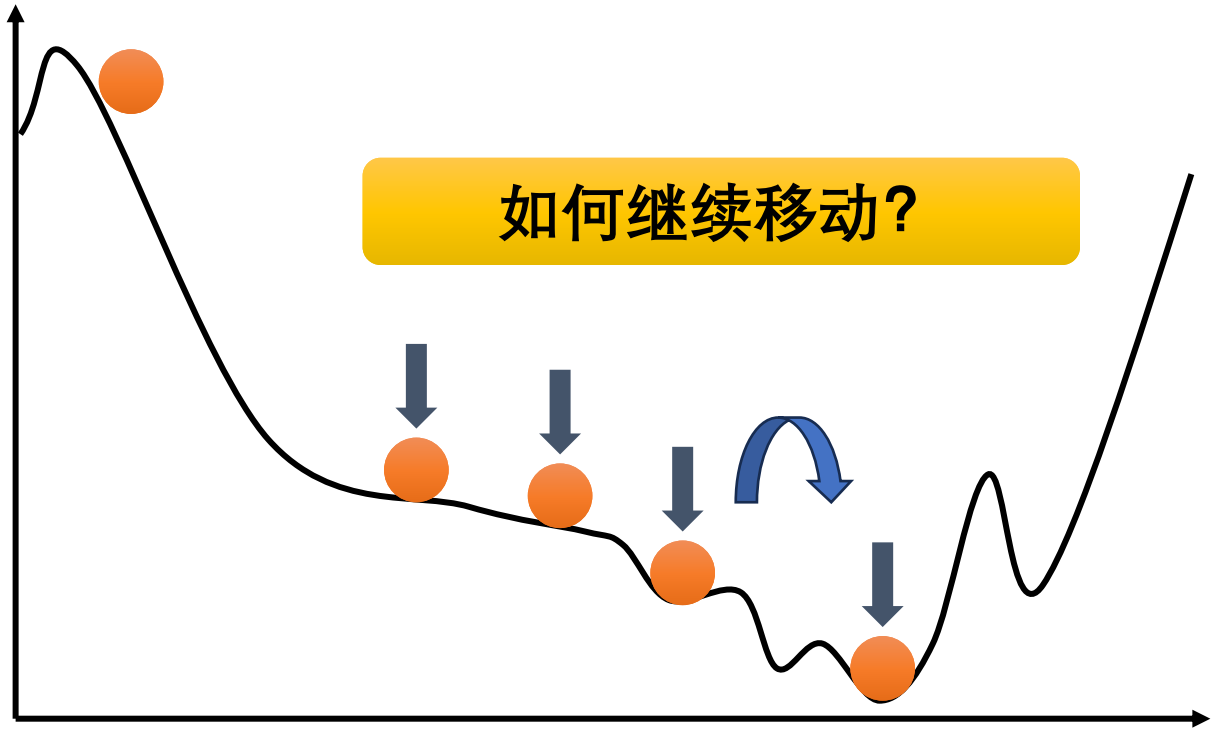

甚至可以从局部最小值
跳到全局最小值?

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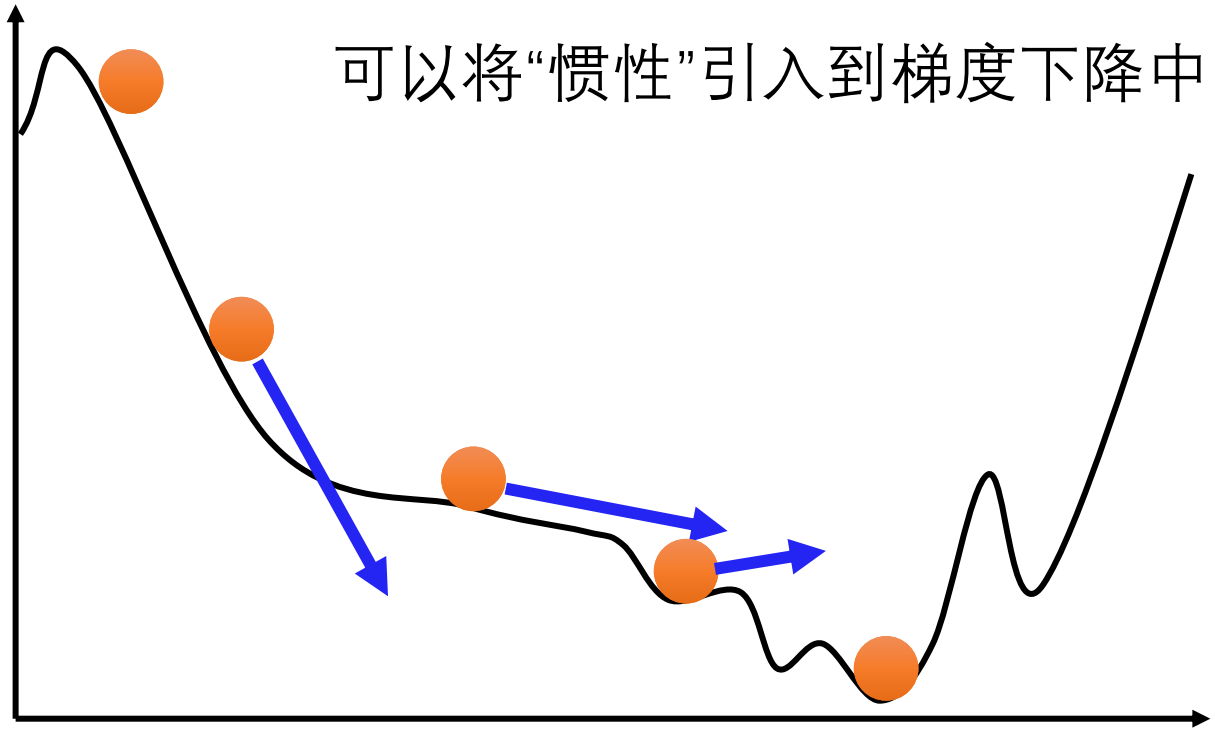


从物理世界角度想象惯性的作用

Adam

Momentum

梯度下降的逻辑:

梯度方向

移动方向

开始于起始位置 θ^0

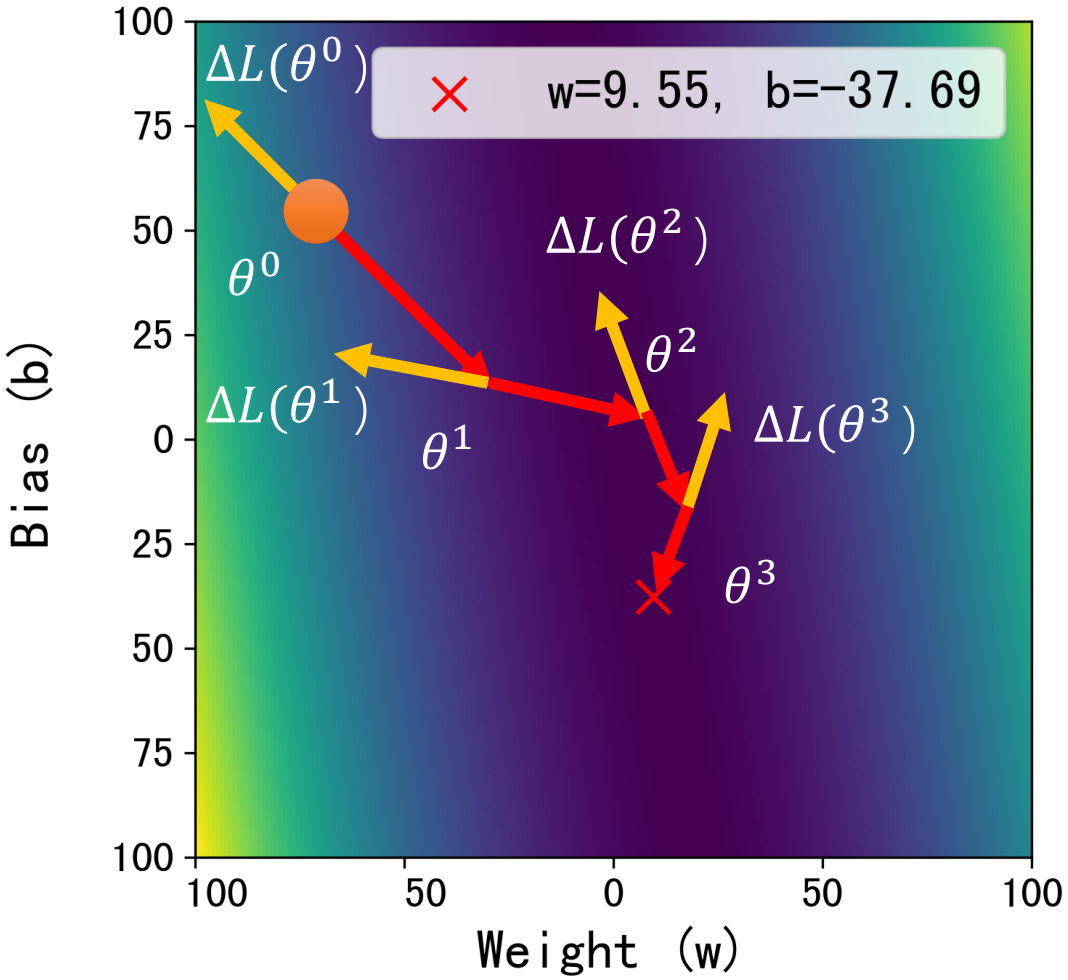
计算θ^0处的梯度$\Delta L(\theta^0)$

移动至$\theta^1 = \theta^0 - \eta\Delta L(\theta^0)$

计算θ^1处的梯度$\Delta L(\theta^1)$

移动至$\theta^2 = \theta^1 - \eta\Delta L(\theta^1)$

⋮



Momentum

梯度下降的逻辑:

梯度方向　惯性方向

移动方向

开始于起始位置 θ^0

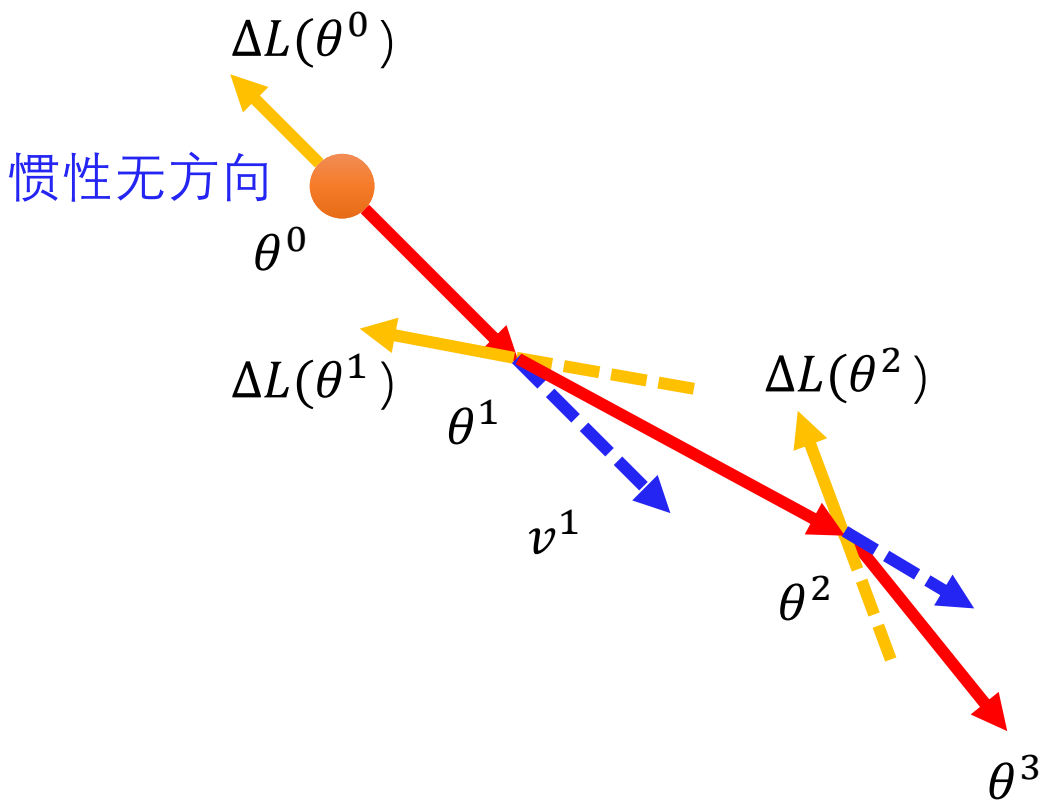
速度方向$v^1 = \lambda v^0 - \eta \Delta L(\theta^0)$

移动至$\theta^1 = \theta^0 + v^1 = \theta^0 - \eta \Delta L(\theta^0)$

计算θ^1处的梯度$\Delta L(\theta^1)$

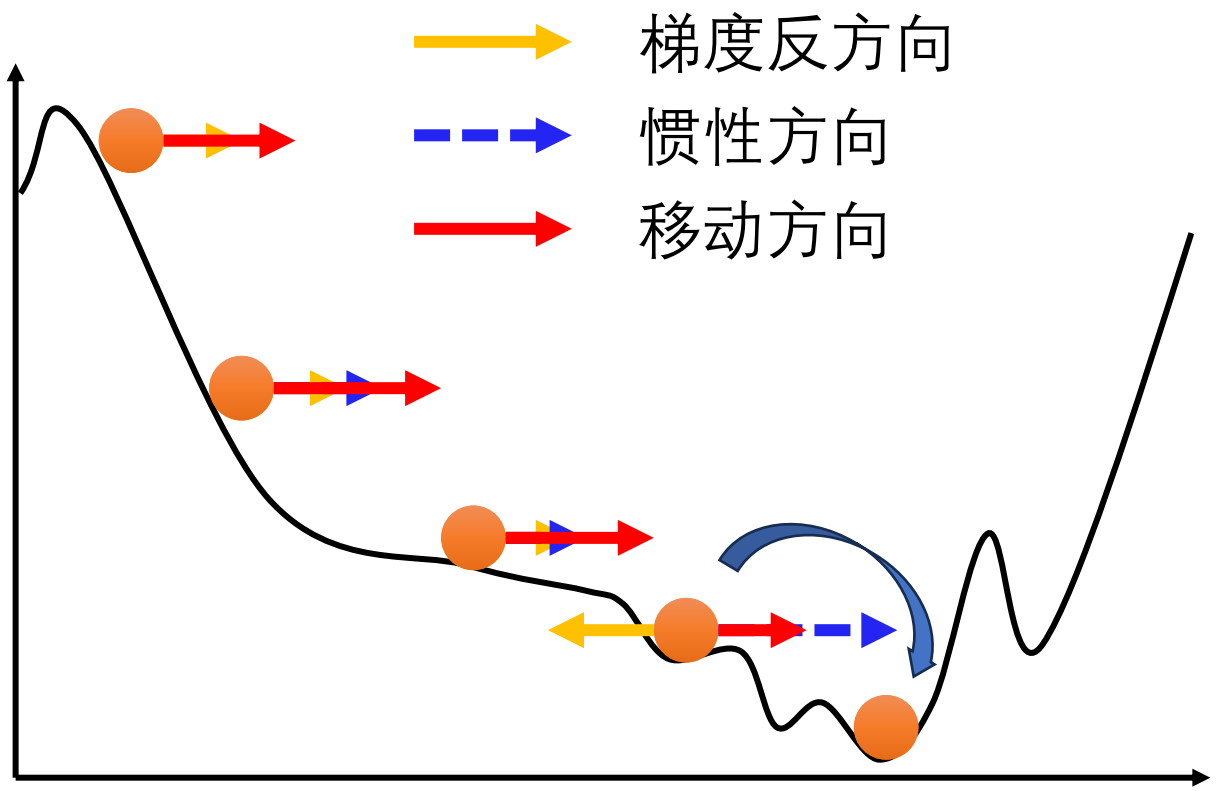
速度方向$v^2 = \lambda v^1 - \eta \Delta L(\theta^1)$

移动至$\theta^2 = \theta^1 + v^2$



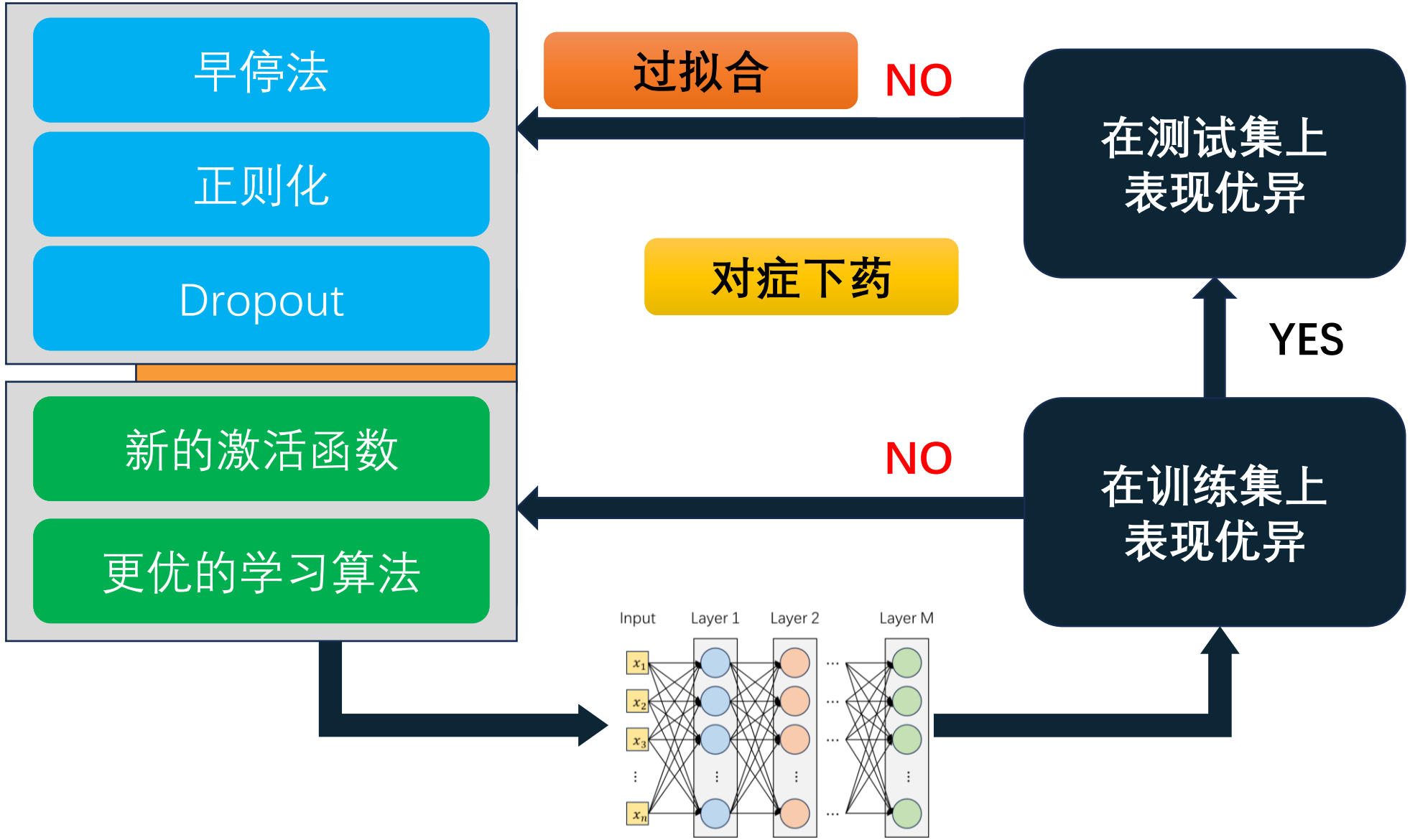
Adam

RMSProp+Momentum



用Momentum的梯度下降

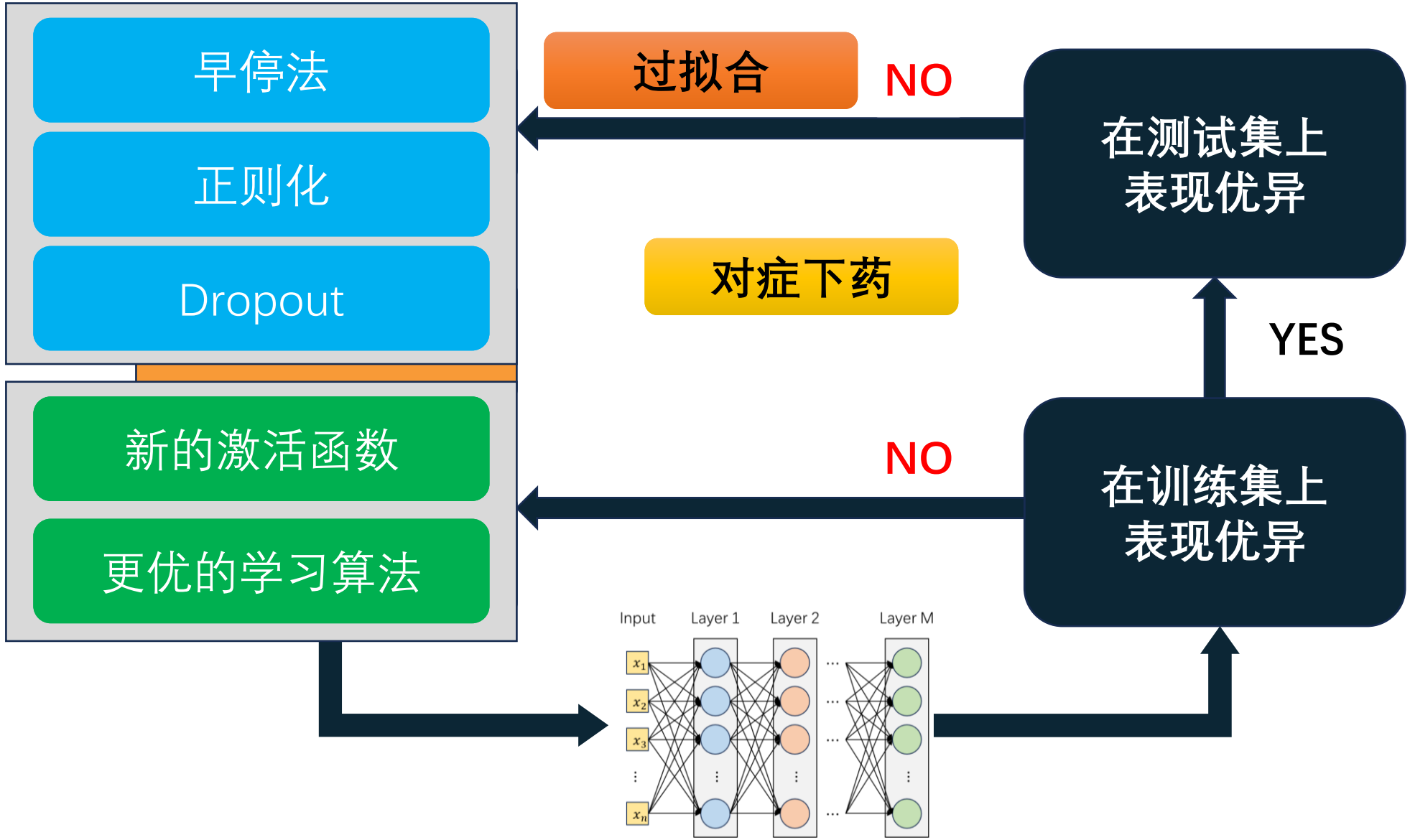
训练深度学习模型的步骤



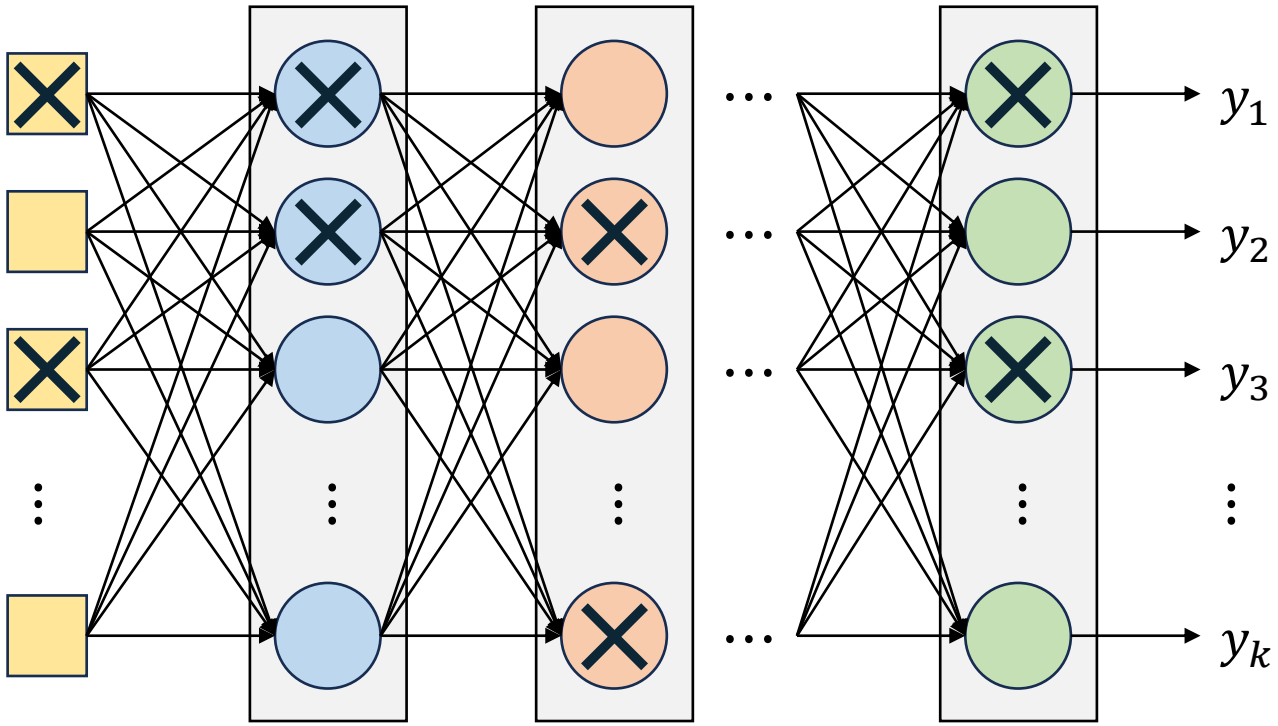
应对过拟合的方法（第二章内容）

- **什么是回归问题?**
  - ✓ 任务预测或者推断
  - ✓ 输出是连续数值
- **机器学习的通用框架**
  - ✓ 训练-测试阶段
  - ✓ 训练阶段包括三个步骤
- **线性回归问题**
- **损失函数&目标函数**
- **梯度下降&学习率选择**
- **模型泛化**
- **过拟合及解决方法**
  - ~~✓ 更简单的模型~~
  - ✓ 数据增强
  - ✓ 正则化
  - ✓ 交叉验证
  - ✓ 早停法
- **欠拟合及解决方法**

训练深度学习模型的步骤



Dropout



Dropout以 $p = 0.5$ 的概率丢弃特征或节点

！！！每次更新都需要重新Dropout！！！

Dropout：为什么有用?

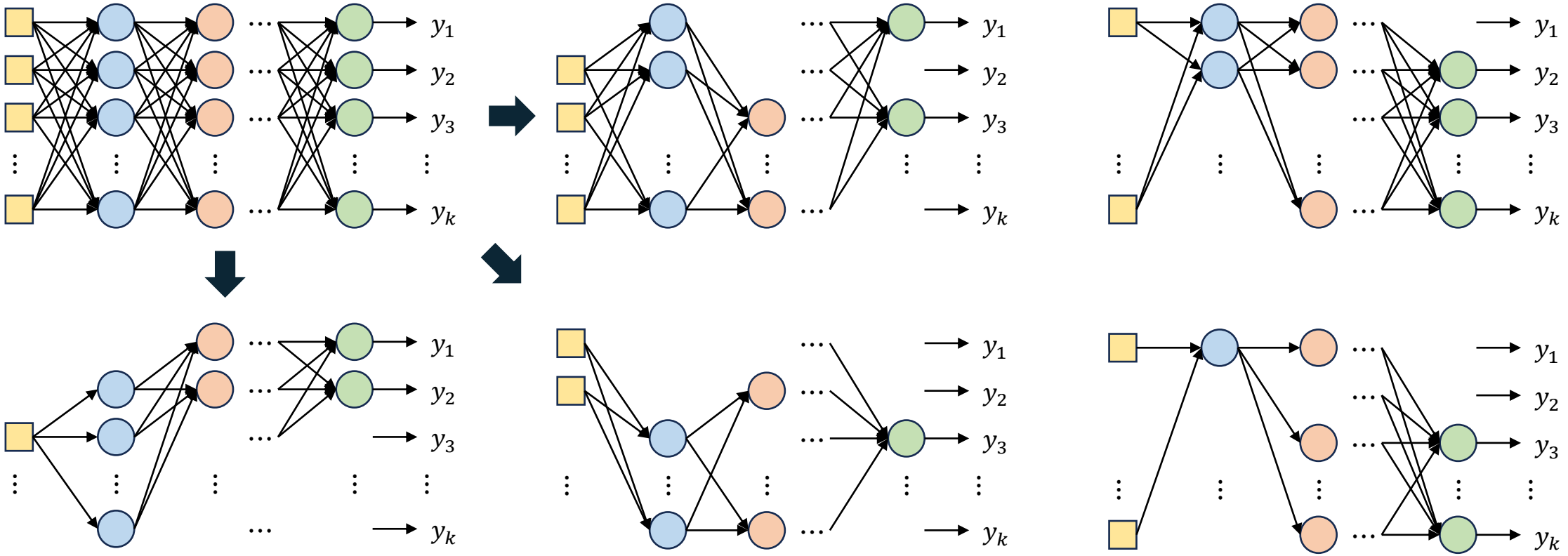
Dropout使网络变得极为简单

每个结点需要“加倍努力”变得更强

！！！每次更新都需要重新Dropout！！！

Dropout：为什么有用?

Dropout相当于训练了极多个不同的简单神经网络

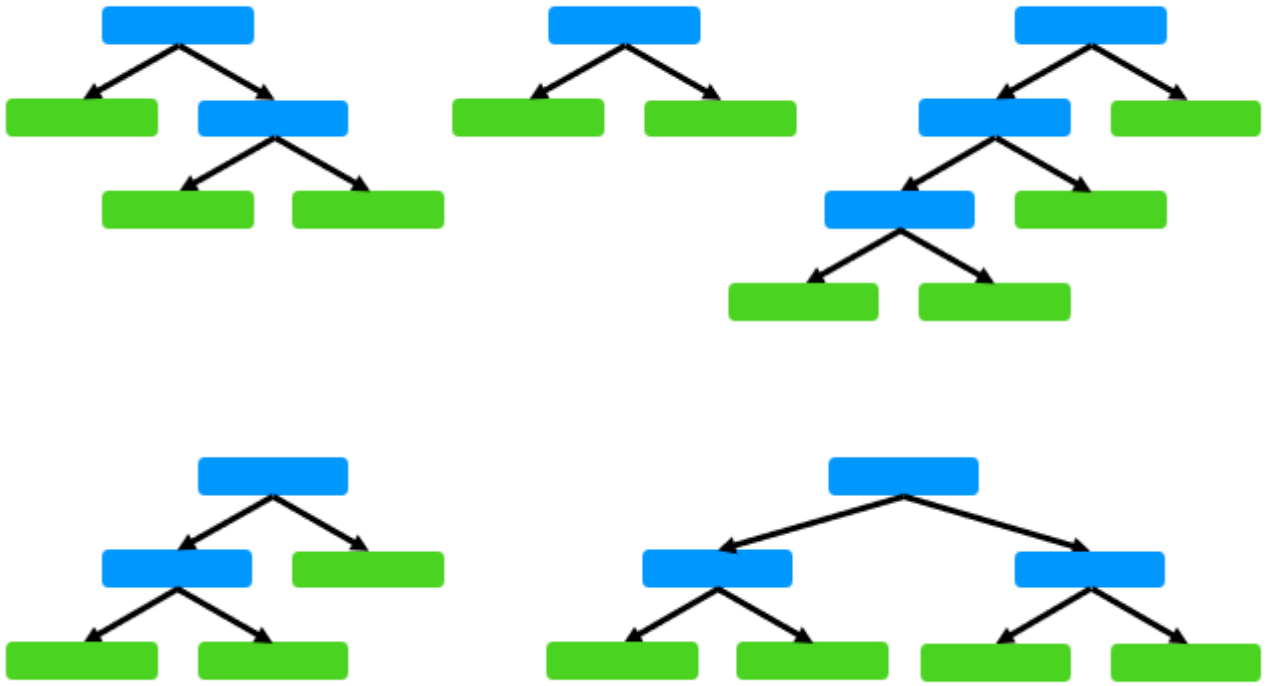


融合多个弱分类器的结果，能够达到极好的效果，并具有更强的泛化能力

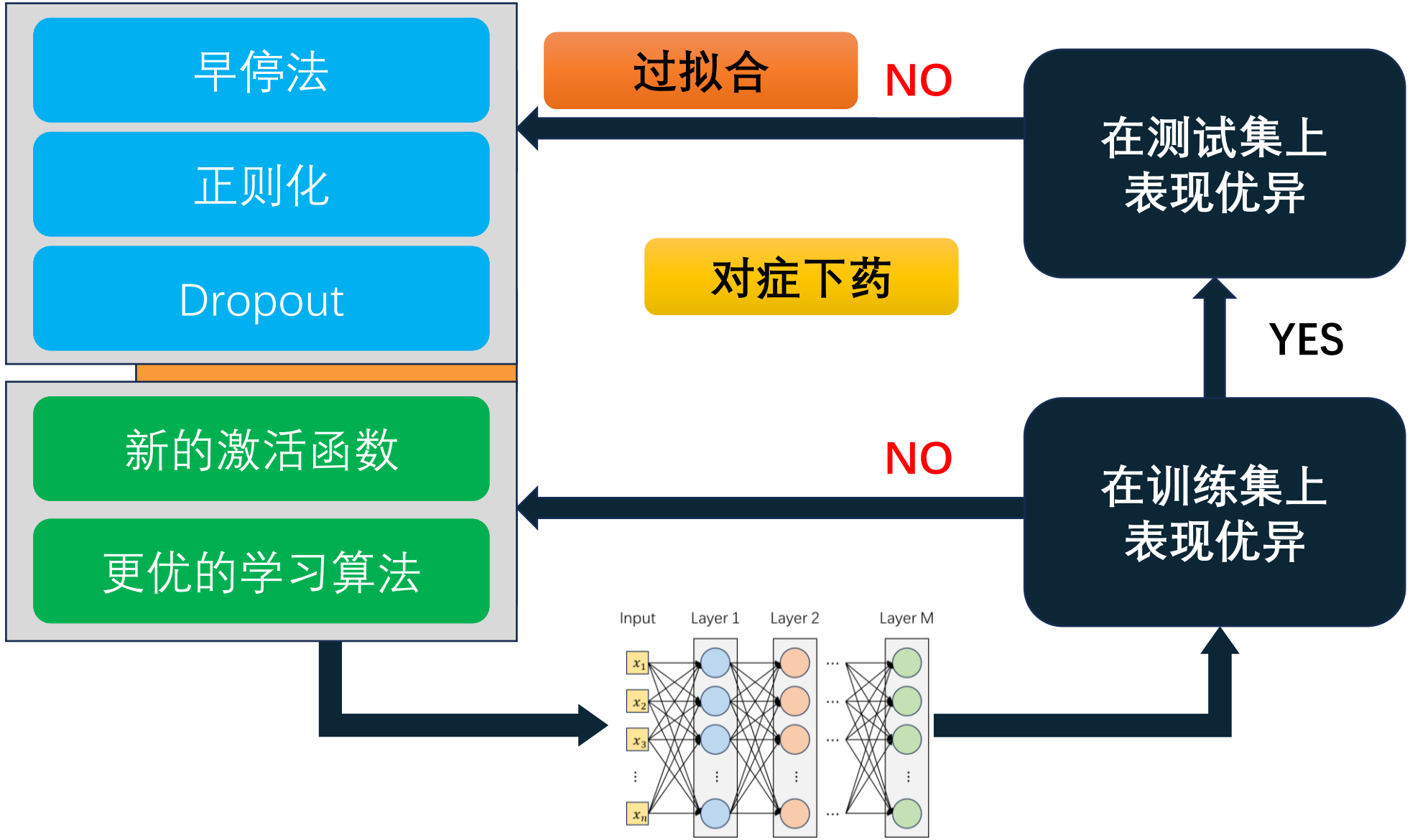
Dropout：为什么有用?

随机森林的优势

- 通过构建多个不同的决策树并投票表决，可以降低过拟合的风险，使得模型更具泛化能力
- 通过随机选取样本和特征，极大的提高了模型的稳定性，减少了单一决策树所带来的随机性



总结


早停法
正则化
Dropout
新的激活函数
更优的学习算法
过拟合
NO
对症下药
在测试集上
表现优异
YES
NO
在训练集上
表现优异
Input
Layer 1
Layer 2
Layer M